

Solution of a system of ODEs (Rachel Batra)

Ques 1: Solve

$$dx/dt = 4 - x^2 - 4y^2$$

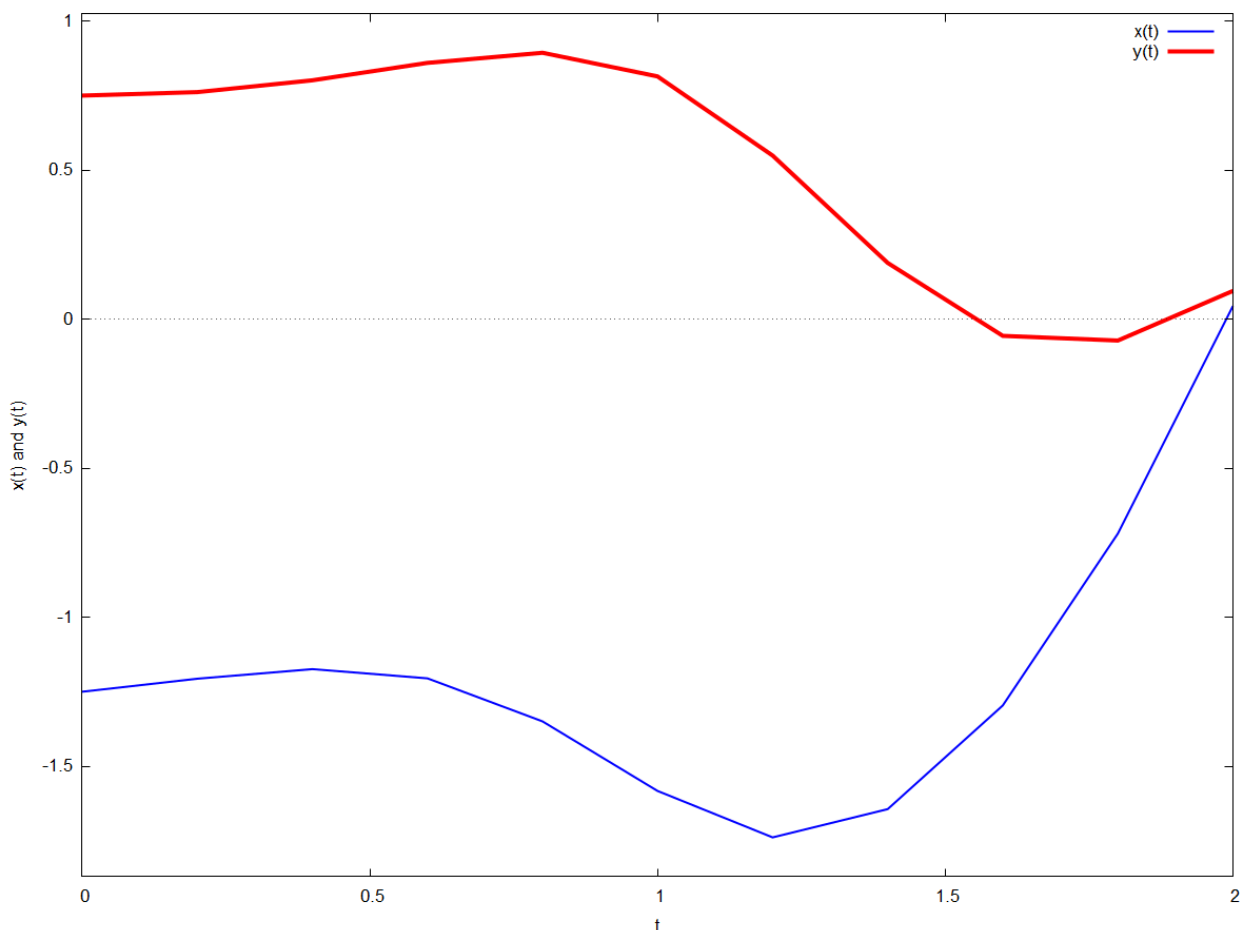
$$dy/dt = y^2 - x^2 + 1$$

where $x(0) = -1.25$ and $y(0) = 0.75$

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-- eqn1 : 'diff( x , t ) = 4 - x ^ 2 - 4 * y ^ 2 ;  
> eqn2 : 'diff( y , t ) = y ^ 2 - x ^ 2 + 1 ;  
list : rk ( [ rhs ( eqn1 ) , rhs ( eqn2 ) ] , [ x , y ] , [ - 1 . 25 , 0 . 75 ] , [ t , 0 , 2 , 0 . 2 ] ) ;  
xList : makelist ( [ list [ i ] [ 1 ] , list [ i ] [ 2 ] ] , i , 1 , length ( list ) ) ;  
yList : makelist ( [ list [ i ] [ 1 ] , list [ i ] [ 3 ] ] , i , 1 , length ( list ) ) ;  
wxplot2d ( [ [ discrete , xList ] , [ discrete , yList ] ] , [ t , 0 , 2 ] , [ style , [ lines , 2 ] , [ lines , 4 ] ] , [  
ylabel , "x(t) and y(t)" ] , [ xlabel , "t" ] , [ legend , "x(t)" , "y(t)" ] ) $
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$$\frac{d}{dt}x = -4y^2 - x^2 + 4$$

$$\frac{d}{dt}y = y^2 - x^2 + 1$$



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The source of this Maxima session can be downloaded [here](#).